Cover Page

- **Problem Statement/Concept Title**

  YatraKavach: landslide and crowd risk early warning for the Vaishno Devi pilgrim route

- **Theme**

  Theme 11: Disaster Mitigation & Resilience

- **Team Name (Registered on portal)**

  kavach

- **Team ID(Registered on portal)**

  HSC|JK|00032

- **State/UT**

  Jammu & Kashmir

The warning existed. The decision did not.

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Problem – Proposed Solution

Core problem

8 Jul 2022 Amarnath cave
Flash flood. No rain data from the upper reaches.

14 Aug 2025 Chasoti, Machail yatra
IMD alerts for the region; the yatra went on. About 61–64 dead.

26 Aug 2025 Adhkuwari, Vaishno Devi
Landslide at 14:40; at least 34 dead. Old track closed by ~12:00.

Each time, a warning reached some official. Nobody turned it into "close this 500 m stretch, move these people, now", and nobody can show who decided. Forecasts are district-scale; crowd systems (RFID, CCTV) have no hazard input.

Proposed solution: YatraKavach

Rain + slope → Segment hazard → + people on segment → Proposed action → Officer approves → CAP alert (en/hi/ur)

Detailed explanation of the proposed solution

- 35 segments (~500 m) on the three Vaishno Devi tracks.
- Hourly landslide hazard from 3 h, 24 h and 15-day rain against published Himalayan thresholds, stepped up on steep slopes.
- People per segment (SIMULATED RFID) and an AI crowd count from a camera frame (DM-Count).

How does it address the problem

- Turns a district warning into one proposal per track and level: hold, clear or reopen.
- A named officer approves; RED needs two officers; escalation after 10 min.
- Approved alert goes out as CAP 1.2 (the SACHET format) in English, Hindi and Urdu.

Innovation and uniqueness of the solution

- Segment hazard × people exposed now, not a district bulletin.
- Models the queue a hold creates: closing a track moves risk to the people below.
- Hash-chained audit log of who decided what, and when.

Sources: Onmanorama, MoES (2022); ETV Bharat (2025); The Tribune (2025). Full list on the References slide.

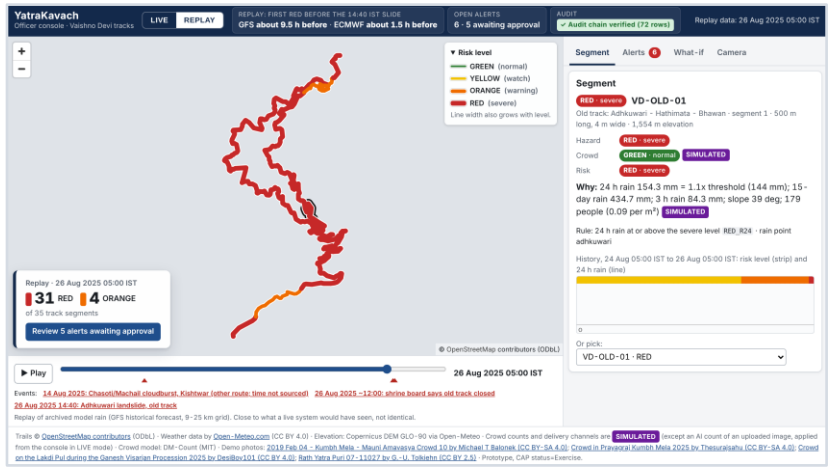
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Technical Approach

Technologies used

- **Backend:** Python 3.12, FastAPI, PostgreSQL 16 + PostGIS 3, pytest (258 tests pass)
- **Console:** MapLibre GL web map, served by the API; Docker Compose to run it all
- **Data:** OpenStreetMap tracks, Copernicus DEM slope, Open-Meteo rain (ECMWF, GFS, ERA5)
- **AI crowd count:** DM-Count (MIT) on ONNX Runtime · Alerts: CAP 1.2 XML

Methodology and process for implementation



Working prototype: replay at 26 Aug 2025 05:00 IST, old track RED with the reasons in numbers

1 Split tracks into segments (OSM + DEM)

2 Score hazard every hour per segment

3 Add people on the segment

4 Propose → approve → CAP alert

5 Validate: replay of 26 Aug 2025

Github/Gitlab link

github.com/RitikBhardwaj-programmer/YatraKavach

Demo Video

JK_kavach_Video.mp4 (4:44), uploaded with this entry; also in the repo: docs/pitch/JK_kavach_Video.mp4

Extent of usage of AI tools

Tool: Claude Code (Anthropic), as coding and research agent.

AI did: research report, most code, tests and docs, an independent code review.

Team did: chose the problem, set requirements, approved every design and fix, validated results.

Share (team estimate): about 60–65% of code and documents AI-generated under the team's direction.

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Impact Potential

Potential impact on the target audience

- Pilgrims on mountain tracks: Vaishno Devi alone allows up to 50,000 a day.
- **Replay of 26 Aug 2025:** old track RED 1.5–9.5 h before the slide, depending on the weather model.
- **If held at the first ORANGE:** 0 pilgrims on the slide segment at 14:00 instead of 674 (SIMULATED).

Benefits of the solution (social, economic, etc.)

- **Social:** earlier, segment-level action to cut disaster deaths (SDG 11.5, SDG 3).
- **Economic:** hold only the segments at risk instead of the whole route (aim, to test in a pilot).
- **Governance:** every alert carries a named approver and time; the log is tamper-evident.

Stakeholders

- Shrine board (SMVDSB) control room, J&K SDMA and district administration
- SDRF and police on duty; IMD and GSI as data sources; NDMA SACHET for delivery
- Pony and palki operators, shopkeepers on the track, pilgrims and families

KPI	Prototype result	Pilot target
Lead time of RED before a slide	1.5–9.5 h (3 models)	≥ 3 h with radar
Alerts with named approver and time	100%	100%
Open alerts at peak (26 Aug, 14:00)	6: one per track and level	≤ 1 per track and level
Pilgrims on slide segment at 14:00 (SIMULATED)	674 → 0 if held at first ORANGE	Decision within 1 h of ORANGE
Camera crowd count error	MAE 60 / image (ShanghaiTech A)	Validated on SMVDSB CCTV

Scale: the same engine works for Amarnath, Machail, Char Dham and Sabarimala; only segments and thresholds change.

Sustainable: open source (MIT), one small server, fits shrine-board control rooms and SDMA. Languages are templates; add Dogri, Kashmiri, Pahari with native speakers.

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Feasibility and Risk Analysis

Analysis of the feasibility of the solution

Working prototype
End to end: hazard, crowd, approval, CAP, audit; 258 tests pass

Open data only
OSM, Copernicus DEM and Open-Meteo; no paid feeds

One small server
Docker Compose; open source (MIT)

Fits existing channels
CAP 1.2 is the format SACHET already uses

Roadmap: 1 Pilot for one monsoon, read-only, with the SMVDSB control room and SDRF → 2 Calibrate with local slide records, IMD radar, 3–5 rain gauges, GSI layer → 3 Integrate RFID, CCTV and approved CAP to SACHET, app and PA.

Potential challenges and risks	Strategies for overcoming these challenges
Rain thresholds come from NH-44 and Nepal studies, not calibrated for these tracks	Read-only pilot for one monsoon; calibrate with local slide records and rain gauges
Forecast rain is on a 9–25 km grid; lead time varies by model (1.5–9.5 h)	Add the IMD radar feed; show the model spread to officers, not one number
Alert fatigue: rules on monsoon rain can fire all season	One alert per track and level: 27 → 6 open alerts in the replay (measured)
A hold moves the risk: 5.4 persons/m^2 at the Katra gate (SIMULATED)	The simulator shows the queue before approval; staggered holds
Crowd data (RFID, CCTV) is SIMULATED today	Data-sharing MoU with the shrine board; label stays until real data flows
Late or wrong decisions	Two officers for RED, escalation after 10 min, break-glass with a reason, audit log

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Team

Team Composition

RB

Ritik Bhardwaj (Team Lead)

B.Tech Information Technology Engineering, 3rd year, NIT Srinagar · solo entry for J&K

Roles

- Problem framing and the J&K focus
- Requirements, architecture and every design decision
- Directing the AI agent (code, research, review) and checking its work
- Validation of the replay results; pitch and demo

Challenges

- **Time:** one person, about four days from idea to submission.
- **Data gap:** the Machail trek is incomplete in OpenStreetMap, so scope was cut to Vaishno Devi (35 segments).
- **Honest replay:** the first version used the slide it predicted; fixed by dating every past-slide factor.
- **Alert flood:** 27 open alerts in the first replay; grouping per track and level cut it to 6.

Any other that the team would like to share about their journey

"The warning existed. The decision did not."

Reading the reports on the 2025 disasters showed the gap was not a better forecast but the step from a warning to an owned, recorded decision. So I built that step, and I show the weak case too: with two of three weather models the lead time is only about 1.5 h.

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References

Details / Links of the reference and research work

Disasters

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Amarnath 2022, MoES: www.moes.gov.in/sites/default/files/RS-in-English-1300-28-07-2022.pdf

Chasoti 2025, ETV Bharat: www.etvbharat.com/en/!bharat/kishtwar-cloudburst-exposes-laxity-in-response-to-weather-warnings-in-jammu-kashmir-enn25082003499

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Existing systems

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Science and models

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Crowd density limits, G. Keith Still: gkstill.com/Support/crowd-flow/fruin/Fruin1.html

Crowd turbulence, Helbing et al.: arxiv.org/abs/physics/0701203

DM-Count crowd model (MIT): github.com/cvlab-stonybrook/DM-Count

Data used (attribution)

Tracks © OpenStreetMap contributors (ODbL): www.openstreetmap.org/copyright

Rain + Copernicus DEM, Open-Meteo (CC BY 4.0): open-meteo.com/en/docs/historical-forecast-api

Project

Code, research notes, replay data: github.com/RitikBhardwaj-programmer/YatraKavach

Full sourced research: docs/research.md and research_notes/ in the repo. Pilgrim counts, holding queues and delivery channels are SIMULATED; all CAP messages are status=Exercise. Demo photos: Wikimedia Commons, authors in the repo.